

25. Plot Two or More Lines with Legends, Widths and Colors

Aim

To write a Python program to plot two or more lines with legends, different line widths, and colors.

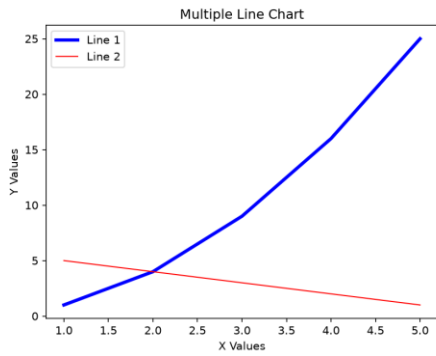
Dataset

X	Line 1	Line 2
1	1	5
2	4	4
3	9	3
4	16	2
5	25	1

Python code:

```
1 import matplotlib.pyplot as plt
2 x = [1, 2, 3, 4, 5]
3 line1 = [1, 4, 9, 16, 25]
4 line2 = [5, 4, 3, 2, 1]
5 plt.plot(x, line1, color="blue", linewidth=3, label="Line 1")
6 plt.plot(x, line2, color="red", linewidth=1, label="Line 2")
7 plt.xlabel("X Values")
8 plt.ylabel("Y Values")
9 plt.title("Multiple Line Chart")
10 plt.legend()
11 plt.show()
```

Output:



Result:

Multiple lines were successfully plotted with different widths, colors, and legends.

26. Create Multiple Plots

Aim

To write a Python program to create multiple plots in a single figure.

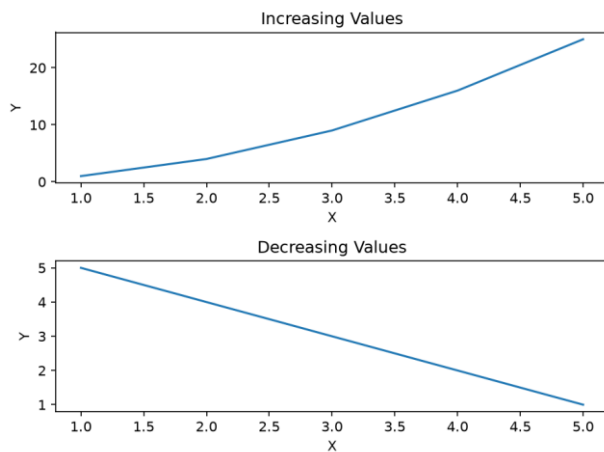
Dataset

X	Y1	Y2
1	1	5
2	4	4
3	9	3
4	16	2
5	25	1

Python code

```
1 import matplotlib.pyplot as plt
2 x = [1, 2, 3, 4, 5]
3 y1 = [1, 4, 9, 16, 25]
4 y2 = [5, 4, 3, 2, 1]
5 plt.subplot(2, 1, 1)
6 plt.plot(x, y1)
7 plt.title("Increasing Values")
8 plt.xlabel("X")
9 plt.ylabel("Y")
10 plt.subplot(2, 1, 2)
11 plt.plot(x, y2)
12 plt.title("Decreasing Values")
13 plt.xlabel("X")
14 plt.ylabel("Y")
15 plt.tight_layout()
16 plt.show()
```

Output



Result

Multiple plots were successfully created using subplot().